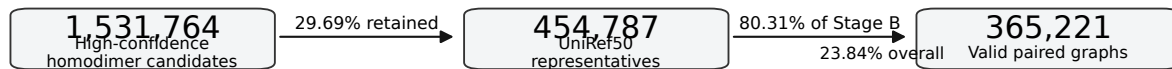
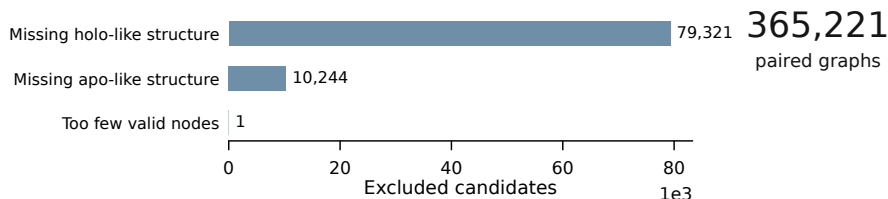


A Dataset construction yield and quality control



exclusion summary

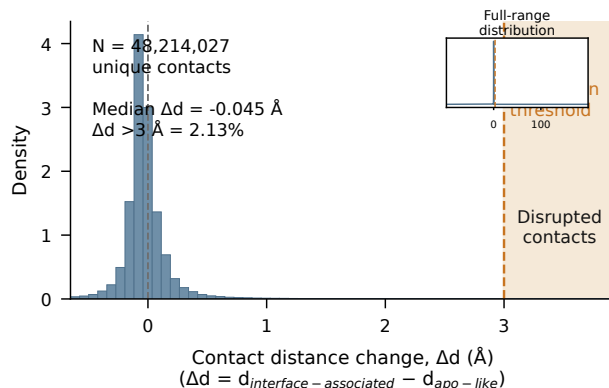
Exclusions after UniRef50 selection, $n = 89,566$



dataset summary

365,221 paired graphs
48.21 M unique apo contacts
1.03 M disrupted contacts
96.43 M directed graph edges

B Distribution of residue-contact distance changes



C Sparsity of contact-level disruption events

C1: Contact-level composition



C2: Per-graph disruption fraction

